ELSEVIER

Tectonophysics 324 (2000) 57–80

TECTONOPHYSICS

www.elsevier.com/locate/tecto

The Variscan belt of Northern France–Southern Belgium: geodynamic implications of new palaeomagnetic data

Emő Márton [a,*], Jean-Louis Mansy [b], Olivier Averbuch [b], László Csontos [c]

[a] *Eötvös Loránd Geophysical Institute of Hungary, Palaeomagnetic Laboratory, Columbus u. 17-23, H-1145 Budapest, Hungary*
[b] *Université Lille I, Sciences de la Terre, Villeneuve d'Ascq, 59655, France*
[c] *Eötvös Loránd University, Geol. Dept., Múzeum krt 4/a, H-1088 Budapest, Hungary*

Received 22 June 1999; accepted for publication 10 April 2000

Abstract

Palaeomagnetic investigations were carried out in Devonian–early Carboniferous rocks of the Variscan foreland chain of Northern France–Southern Belgium in order to reveal the origin of its arcuate shape. The Brabant Parautochthon was sampled in the Boulonnais (near Calais) and near Tournai, while the Ardenne Allochthon was sampled near Maubeuge and in the Givet area. All the sampled localities yielded characteristic remanent magnetization as a result of stepwise demagnetization and component analysis. Fold or tilt tests were possible for three localities, with negative results indicating pervasive remagnetization. The tectonic position was sub-horizontal at two localities, while the tilt was monoclinal for the rest. Therefore, the acquisition time of the magnetic signals was estimated by comparing the palaeolatitude computed from each magnetic component to the palaeolatitudes of Variscan Europe calculated after Van der Voo (1993).

Three components showing: A, a southern B, a near-Equatorial, and C, a northern palaeolatitude are recognized from our data. Since a pre-Variscan age of component A (observed only in Boulonnais, at 10 sites) is not supported by data, it is assigned to an early phase of deformation. Component B (16 sites) was acquired during the peak of the Variscan tectonics (late Westphalian), while component C (five sites) originated during Permian times.

Regardless of the palaeolatitudes, declinations fall between 190 and 210°, thus being conformable with the declinations expected for Variscan Europe. The declinations show no correlation with the arcuate shape of the belt, neither are they different in the Paraauthochthon and in the Allochthon, nor in the different components.

Arc formation by moulding of the Allochthon on the Brabant Parautochthon is, therefore, not supported by these data, since this mechanism requires substantial (opposed) rotations on both wings of the arc. The available palaeomagnetic data are conformable with a pre-formed arc, simply docking to the Brabant obstacle of similar shape. Variable offsets along a main thrust could support a third model, which slightly unfolds the former passive arc.
© 2000 Elsevier Science B.V. All rights reserved.

Keywords: geodynamic implication; palaeomagnetism; Palaeozoic; Variscan belt

* Corresponding author. Tel.: +36-319-3203; fax: +36-319-2303.
E-mail address: h11000mar@ella.hu (E. Márton)

0040-1951/00/$ - see front matter © 2000 Elsevier Science B.V. All rights reserved.
PII: S0040-1951(00)00109-8

1. Introduction

The northern front of the Variscan orogen in central and western Europe is conventionally located in the Rhenohercynian tectono-metamorphic zone (Kossmat, 1927). This zone was first defined from field studies in the Harz, Rhenish and Ardenne massifs and subsequently extented westwards to the British Isles by several authors (e.g., Matthews, 1984a,b; Franke and Engel, 1986; Matte, 1986; Bois et al., 1994). The Ardenne massif is classically divided into two main units separated by a major fault, the Midi Fault (Gosselet, 1888), which overthrusts the Coal Basin (Figs. 1 and 2). North of this fault, the Brabant Massif and its cover crop out in the Boulonnais, Tournai and Coal Basin areas. They constitute the weakly deformed Brabant Parautochthon formerly called the Namur syncline (Fourmarier, 1922). South of the Midi Fault, the more intensely deformed Ardenne Allochthon (Meilliez and Mansy, 1990; Meilliez et al., 1991) crops out in the Avesnois region and near the Meuse valley, corresponding broadly to the formerly defined Dinant synclinorium (Fourmarier, 1922).

Displacement on the Midi Fault varies along the strike, from a few kilometres near Aachen and Liège (Winterfeld and Walter, 1993; Laloux et al., 1997) to more than 50 km in the central part of the Coal basin (Raoult and Meilliez, 1987; Mansy et al., 1997). The main shortening deformation began in Late Visean in the Ardenne Allochthon and in the Late Westphalian time in the Brabant Parautochthon.

The southern allochthonous elements record low-grade (higher epizone) metamorphism with

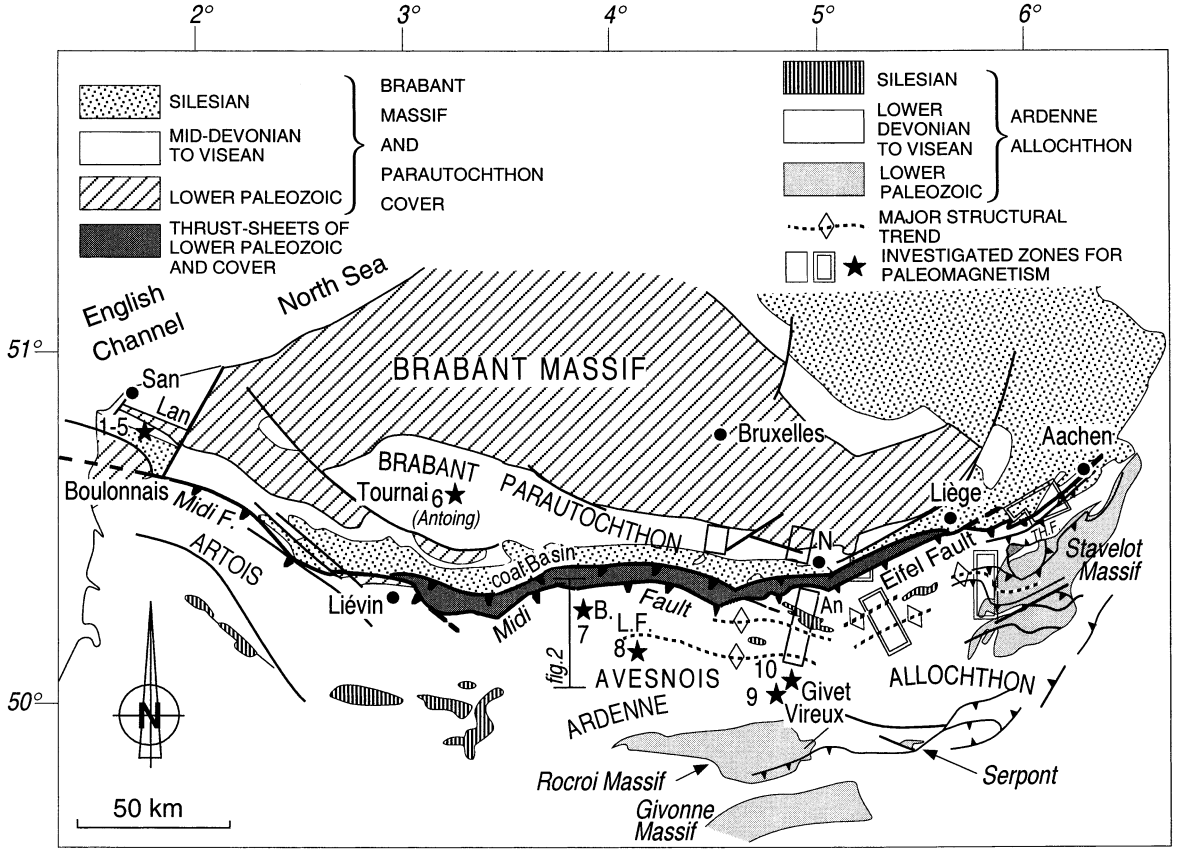


Fig. 1. Location map and structural outlines of the study area.

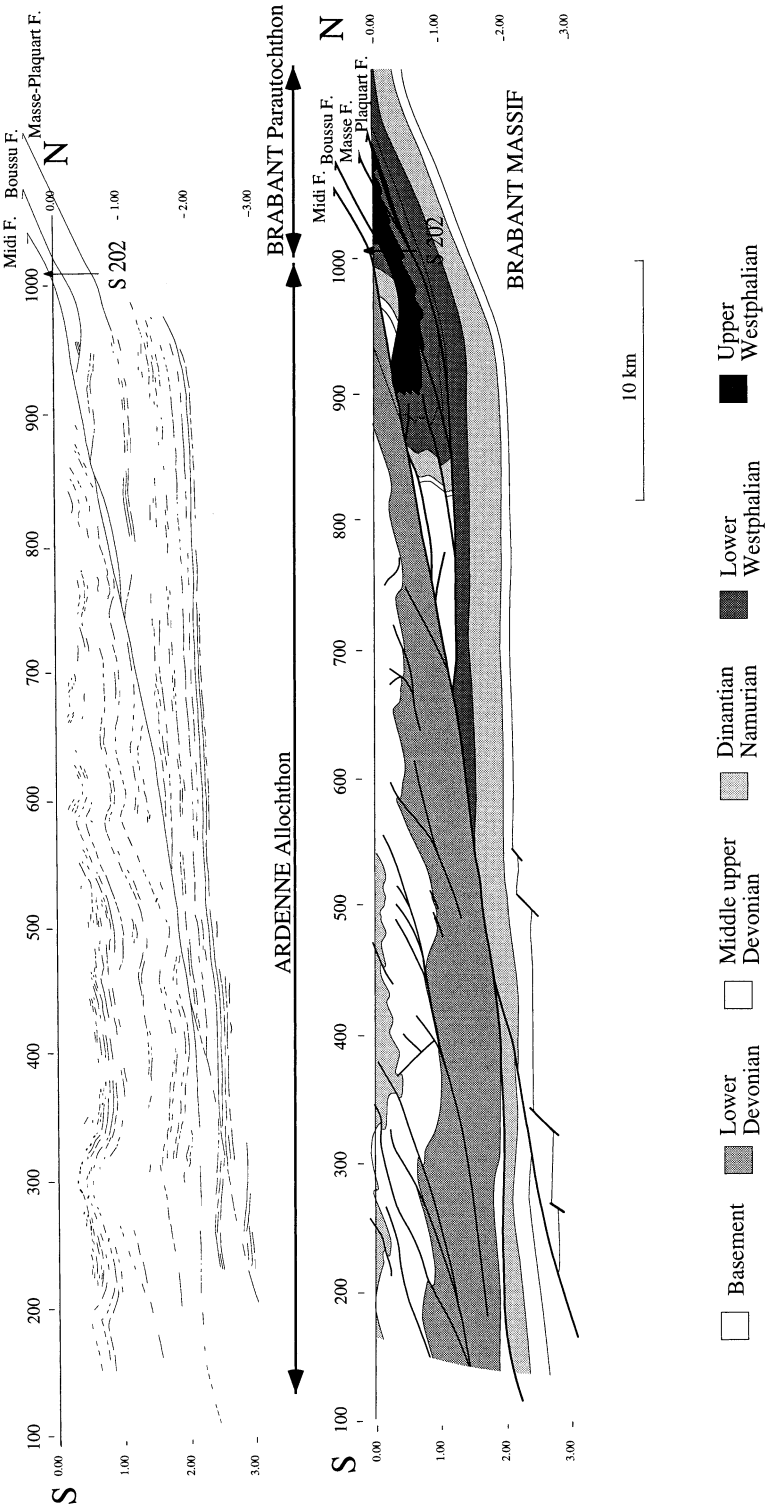


Fig. 2. Schematic cross-section of the central part of the study area based on seismic profiles (after Lacquement et al., 1999).

temperatures not exceeding 400°C (Potdevin and Goffette, 1991; Chamley et al., 1997), while the localities sampled in this study show temperatures that did not exceed 300°C. The Parautochthon records only diagenetic conditions (Fielitz and Mansy, 1999).

The Boulonnais-Ardenne segment of the Variscan chain is classically described as a bent orogen, with structural trends of major fold axes varying from N110° in the Boulonnais to N70° in the Ardenne. Based on palaeomagnetic data, this has already been interpreted as an arched orogen by Perroud (1986), Edel and Coulon (1986), and Thominski et al. (1993), though observations were made only in the eastern part of the Ardenne.

This paper presents the first palaeomagnetic observations from the Boulonnais and new data from the Tournai, Avesnois and Meuse valley regions. Combined with the previously published palaeomagnetic and structural data, an attempt is made to relate palaeomagnetic events to the structural evolution and general kinematics of the chain.

2. Geological overview

2.1. Brabant Parautochthon

The basement (Brabant Massif) is composed of Early Palaeozoic rocks, folded and metamorphosed in the early Devonian (Acadian time). To the northeast and to the south, it is bounded by laterally variable continental to marine sediments of Givetian to Dinantian age. Early Devonian deposits are lacking in most regions, whereas Middle Devonian clastic and Dinantian carbonate sediments form a sedimentary cover with few gaps and increasing thicknesses towards the borders of the Massif (Bless et al., 1980; Muchez, 1989; Meilliez et al., 1991). From the Boulonnais (in France) to the Aachen area (in Germany), a succession of Namurian to Westphalian coal-bearing clastic sequence represents the foreland coal basin of the northwards migrating deformational front. In all exposed areas, cleavage is either very weak or absent.

The WNW–ESE-oriented Boulonnais area (Fig. 3) is situated in the SW margin of the Brabant unit along the eastern side of the English Channel, as an extension of the Variscan structures of Southern England. The Boulonnais region presently represents a faulted inverted structure involving, in addition, Jurassic and Upper Cretaceous sedimentary rocks. Within the core of the massif, there are unmetamorphosed Devono-Carboniferous sedimentary rocks lying unconformably on Silurian shales. The whole sequence was folded and thrusted after mid-Westphalian times. The cover-basement succession can be divided into three structural subunits: the lowest and northernmost Ferques, the intermediate Haut Banc, and the uppermost and southernmost Hydrequent units (Figs. 3 and 4). The style of the NNE vergent folds and thrusts faults (for example Averbuch and Mansy, 1998) shows that the Boulonnais Palaeozoic area forms the westernmost extension of the French Variscan thrust front, which is well constrained in the Nord-Pas de Calais coal basin (Bouroz et al., 1962). The major thrust zone, the Midi Fault, in classical interpretations, lies a few kilometres south of the exposed Palaeozoic strata. The Boulonnais Palaeozoic can thus be considered as the deformed footwall of this major Variscan thrust. In the Boulonnais, the Midi Fault is a complex, rather than a singular, fault zone, and it does not represent the emergent front of the Variscan orogen. Some of the Variscan faults are reactivated during Mesozoic times (e.g., Landrethun Fault, Figs. 1, 3, and 4), while others are sealed by undisturbed Middle Jurassic deposits (e.g., Ferques Fault, Fig. 3., Mansy et al., in press).

The Tournai Palaeozoic massif is situated in a more intermediate section of the Northern France–Southern Belgium Variscan front with a general E–W trend (see Fig. 1), some tens of kilometres north of the major Midi Fault Zone. It is dominantly a Carboniferous carbonate cover affected by very gentle folding and contractional faults with little offset (Hennebert, 1998).

The Coal Basin crops out south of the Tournai area with a thick and in places intensely deformed Namuro-Westphalian coal sequence. The onset of shortening was characterized by the development of metric to kilometric ramps and contractional faults that became folded as deformation proceeded (Humblet, 1941; Chalard, 1945).

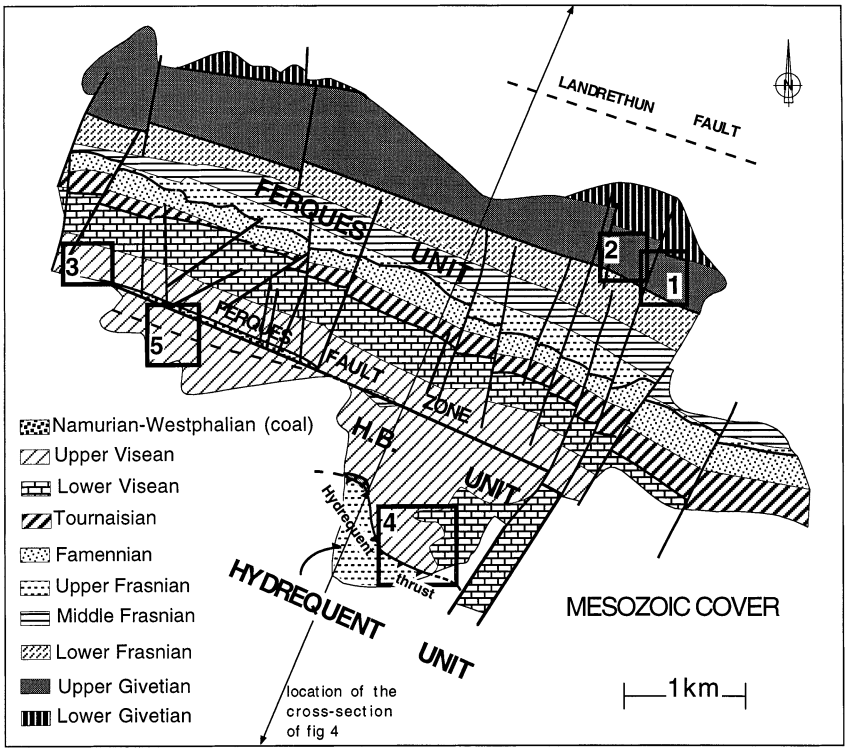


Fig. 3. Location map and simplified geological map of the Boulonnais area (after Mansy et al, in press).

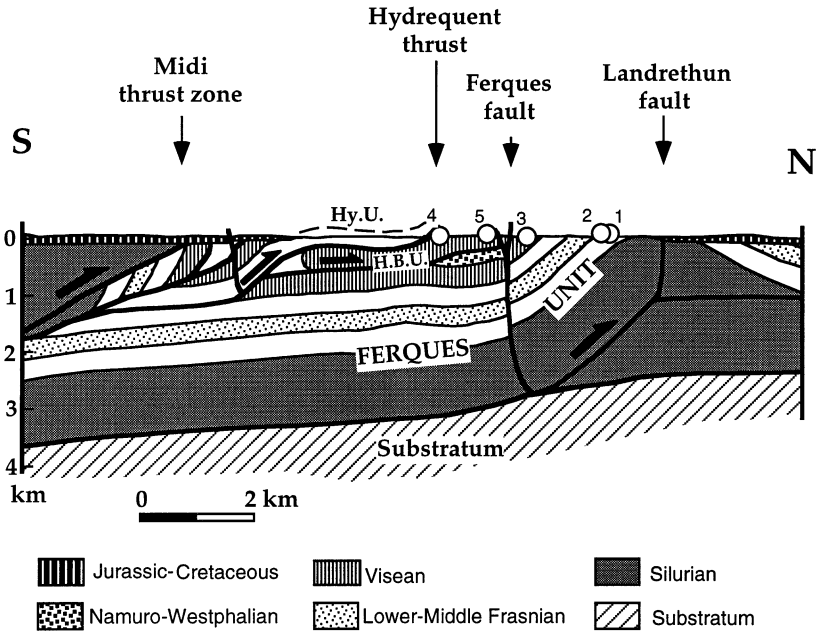


Fig. 4. Cross-section of the Boulonnais area (after Averbuch and Mansy, 1998); note the structural position of the sampled sites.

Superimposed on these early folded and faulted structures is a succession of out-of-sequence thrust faults that often have kilometric displacements (Mansy et al., 1997; Hance et al., 1999). The deformation in the Coal Basin suggests that the Variscan front lies several kilometres to the north.

2.2. Ardenne Allochthon

In the Rocroi, Givonnne and Stavelot massifs (Fig. 1), parts of the Ardenne Allochthon, Lower Devonian strata unconformably overlie strongly folded lower Palaeozoic rocks.

The Lower to Middle Devonian cover lies unconformably and without décollement or detachment on this basement (Meilliez et al., 1994). It consists of a fining-upwards clastic group deposited in a continental to marine environment. The Middle Devonian rocks accumulated on a carbonate platform grading southwards into a slope. In Late Devonian times, a strongly differentiated clastic and carbonate shelf developed. This was succeeded by a condensed carbonate platform sequence in Early Carboniferous times. The late Early Carboniferous is characterized by strongly differentiated facies distribution. The Late Carboniferous clastics are interpreted as a basin fill that developed in front of the emerging Variscan nappes.

The Variscan shortening gave rise to a system of folds and faults, which record a progressive deformation (Mansy and Meilliez, 1993; Mansy et al., 1995). Irrespective of the scale, the shortening is expressed as asymmetrical north verging folds with short steep limbs commonly cut by flat to shallow dipping faults. In the whole Allochthon, a penetrative cleavage is observed, which affects both the Lower Palaeozoic and its unconformable Devonian–Carboniferous cover. In southern parts, the cleavage is post-folding. In some places, a second crenulation cleavage can be observed. In the central part of the Ardenne Allochthon, tectonic transport is directed northward, although in detail, variations in fold trends and cleavage strike orientations from N70 to N120° are found.

The basement and the cover of the southern margin of the Rocroi Massif record an epizonal metamorphism that decreases in intensity both to the north and to the south. Ages obtained from chlorite flakes constituting the schistosity range from 315 Ma in the south to 290 Ma in the north (Piqué et al., 1984), similar to the trend observed in Germany (Ahrendt et al., 1983). The general northwards younging of deformation is also corroborated by Uppermost Visean sedimentological features such as slumps and megabreccias.

In the southern part of the Ardenne Allochthon, illite crystallinity indicates anchizonal burial. In some places, clay assemblages show that burial and metamorphism were weak (Chamley et al., 1997). This observation suggests that Devonian rocks had only a moderate overburden (2–4 km) and that they were rapidly uplifted during Late Visean–Early Namurian times (Mansy et al., 1995). Helsen (1992, 1995) has measured CAI values between 3.0 and 3.5 (i.e., 100–150°C) on the northern flank of the Rocroi massif, and values of 4.5–5.0 (250–300°C) have been obtained near the Stavelot Massif (Fielitz and Mansy, 1999).

The Avesnois region is located to the NW of the Rocroi Massif in the area affected mainly by diagenesis to anchizonal metamorphism. It is composed of different north-vergent, internally deformed thrust-sheets, involving 3 km thick Lower Devonian to Visean sequence (Khatir, 1990). The sampled Givet region of the Meuse valley is located to the north of the Rocroi massif and has the same structural and diagenetic–metamorphic conditions as the Avesnois region.

3. Palaeomagnetic sampling

In the framework of a French–Hungarian Intergovernmental Project (Balaton Project F-45/96), a total of 144 samples were drilled at 10 Palaeozoic localities (Fig. 1, Table 1) and oriented in situ by magnetic compass during two field campaigns in 1996–1997. The sampling localities are within the exposed NW, central and eastern parts of the Variscan chain of Northern France–Southern Belgium, namely the Boulonnais (five localities), the Tournaisis (one locality), the Avesnois (two localities) and the Ardennes-Meuse valley (two localities) areas. Sampling was not extended east of the Givet region since recent

Table 1
List of the palaeomagnetic localities

	Locality	Latitude	Longitude	Magnetic declination	Age
Boulonnais - Ferques Unit					
1	Griset	50°50′00	1°47′30	−2.9°	Upper Givetian: 374–380 Ma
2	Banc Noir	50°50′00	1°47′30	−2.9°	Upper Givetian: 374–380 Ma
3	Leulinghen	50°48′30	1°46′00	−3.0°	Visean: 333–352 Ma
Boulonnais - Haut-Banc Unit					
4	Basse-Normandie	50°48′00	1°46′30	−3.0°	Visean: 333–352 Ma
5	Moines	50°48′30	1°46′00	−3.0°	Visean: 333–352 Ma
Tournaisis					
6	Antoing	50°37′00	3°20′00	−2.0°	Tournaisian: 352–360 Ma
Avesnois					
7	Bettrechies	50°13′20	3°43′00	−2.1°	Givetian: 374–380 Ma
8	Limont-Fontaine	50°12′00	3°48′00	−2.0°	Visean: 333–352 Ma
Ardennes					
9	Vireux	50° 5′00	4°42′30	−1.7°	Lower Emsian: 387–394 Ma
10	Givet	50°7′30	4°50′00	−1.7°	Givetian: 374–380 Ma

palaeomagnetic data are available for this part of the Ardennes (Thominski et al., 1993; Molina Garza and Zijderveld, 1996).

3.1. Brabant Parautochthon

3.1.1. Boulonnais region

There are several quarries in the Palaeozoic of Boulonnais. Most of them were visited, but only a few appeared to be suitable for palaeomagnetic investigation. Three localities, Griset, Banc Noir and Leulinghen quarries (respectively points 1, 2 and 3 in Fig. 3) of Upper Givetian and Visean age, have been sampled in the lowermost thrust unit known as the Ferques Unit (see the structural positions of the sites in Fig. 4).

In Griset, a thick sequence of dark grey Givetian limestones and shales is quarried. Locally, beds rich in corals and other reef-building organisms can be found, especially towards the top of the exposed series. In the quarry, three sites were sampled with different lithologies and in different stratigraphic positions: dark grey marl, black marl, black shale-silt and silty limestone. The sites represent about 100 m in stratigraphic thickness.

In the Banc Noir quarry, a Givetian dark limestone–shale sequence is exposed. A green index horizon was sampled, consisting of clays, which could be of volcanic origin, and a reddish marl intercalated with the black–grey limestones.

In the Leulinghen quarry, the sampled material comes from a part of the quarry lying at a considerable distance from the Ferques fault. The Early Visean rock consists of cycles of dark, 0.5–1 m thick dolomitic limestone beds topped generally by a few centimetres of thin reddish, marly-clayey material. The latter may be of palaeo-soil or tuff origin. The sampled rocks are reddish palaeo-soils with dolomitic debris and dark dolomitic beds with small pebbles. The facies is interpreted as a shallow carbonate platform.

Two other localities [Basse-Normandie (point 4 in Fig. 3) and Moines quarries (point 5 in Fig. 3)] have been sampled within the Visean limestones–dolostones of the intermediate Haut-Banc unit (see Table 1). In the Basse Normandie quarry, the wall near the railroad cut was sampled, where a fold and ramp is exposed. The structure has been analysed by Cooper et al. (1983), Averbuch and Mansy (1998). The fold was not accessible, and samples were collected from the footwall of the large ramp, 4 m beneath the duplex horizon. Here, the beds were monoclinal with a southerly dip and consisted of metre-thick light grey limestone beds, that alternate with thin marly–sandy layers. Some of these have oxidized to a secondary red colour.

There are also some clayey beds that may also have an original red colour. Several sites were sampled in red clays–shales, in dark grey–brownish siltstones and dark, thin limestone intercalations.

In the Moines quarry, Early Visean limestones intercalated with thin marls or shales are exposed. Thin intercalations of limestones and fine-grained sandstones–shales, all dark grey in colour, were sampled.

3.1.2. Tournai area

In the Tournai area, one locality, the Antoing quarry, has been investigated (point 6 in Fig. 1), where subhorizontal black Tournaisian marly limestones are exploited. The quarry yards are cross-cut by major faults, and the rigid material is also frequently fractured. The samples are from the eastern yard. The black, often fossiliferous, rock is composed of thick limestone and thin shale layers. Both marly limestone and thinner shale intervals were sampled.

3.2. Ardenne Allochthon

3.2.1. Avesnois area

Two folded sequences have been investigated: one in the Givetian marly limestones, in the Bettrechies syncline lying a few kilometres south of the Midi Fault (point 7, in Fig. 1; see also Fig. 2), and the other from the synclinally folded Visean limestones in the Avesnes thrust unit (Limont-Fontaine quarry, point 8, in Fig. 1).

In the Bettrechies quarry, the Late Devonian dark carbonate consists of regular alternations of 30–80 cm thick limestones, marly limestones and 5–10 cm thick black shales, marls. Both rocks can be fossiliferous. Penetrative or weaker cleavage, depending on clay content, affects both rocks. Frequently, the limestone beds are cross-cut by white calcite veins, and both rock varieties are also fractured. Layer-parallel slip, sometimes filled by calcite veins, is also very frequently observed, especially within thicker shale horizons. The quarry exposes a kilometric syncline, the southern limb of which is more deformed, than the northern limb (Mansy et al. 1995). The structure is truncated on the top by subhorizontal Cenomanian and younger beds. Seven sites were drilled on the two limbs of the Palaeozoic syncline in sectors that seemed less deformed and farther from detachments. Limestones, marly limestones and dark shales were collected.

This quarry is especially valuable, since geotemperature studies were carried out, using different methods. Fluid inclusions in the calcite veins (Kenis et al., 2000) suggest a stepwise decrease in temperature from 300 to 70°C that is related to different tectonic events. Vitrinite reflectance values (Degardin, pers. commun.) range between 4 and 3.5 and indicate the same range of temperature. The CAI (Conodont Alteration Index) values (Helsen 1995) are between 5 and 4.5, corresponding to temperatures of 300 and 250°C.

The Limont-Fontaine quarry exploits mid-Visean limestones. The dark carbonate consists of a cyclic alternation of thicker limestone layers, often dolomitized, and thinner, sometimes reddish marly–shaley horizons, interpreted as palaeosoils. Hectometric slumps are frequent. Smaller limestone beds immediately below the red clays may contain intraclasts or brecciated layers, while the bulk of the limestone is composed of micrite. No cleavage is apparent in the sampled rocks. The quarry exposes a syncline, two flanks of which were sampled.

3.2.2. Meuse valley, Givet area

Two localities were investigated in the region of Givet in anchi-metamorphosed Devonian rocks of the northern limb of the Rocroi massif (Fig. 1): one in the north-dipping Lower Emsian sandstones of the Vireux quarry and another in subvertical Givetian limestones in the vicinity of Givet.

In the Vireux quarry, the rocks are thick, brownish, massive sandstones, often showing ripple marks, and thinner, very compact, black shales. A strong cleavage is observed in the latter. Dark shales in several beds were sampled, but in the same structural position, with a northerly dip.

The locality near Givet is in a road cut on the eastern flank of the Meuse, near a small bridge, on the French side. The sampled rock was a Givetian black, sometimes dolomitic limestone in 20–10 cm thick beds, with a few intercalations of clay and fine-grained limestone. The sequence is

cut by calcite veins and cracks. Two of the thin dark limestone beds from the steep (75°) overturned limb were sampled.

4. Rock magnetic properties and palaeomagnetic analysis

The standard-sized specimens cut from the drill-cores were subjected to magnetic susceptibility anisotropy (AMS) measurements in Lille (KLY-2 Kappabridge), NRM (Natural Remanent Magnetization) measurements and demagnetization in increments by an AF (alternating field) or thermal method or by the combination of the two in Budapest (JR-4 magnetometers, two axes Cryogenic magnetometer, Schonstedt oven, Schonstedt AF demagnetizer and AF demagnetizer built at the Technical University, Budapest peak field 0.24T). Magnetic minerals were also identified in Budapest by IRM (Isothermal Remanent Magnetization) acquisition characteristics (Pulse Magnetiser, Molspin) and by stepwise thermal demagnetization of the three-component IRM (Lowrie, 1990).

4.1. Intensity of the magnetic signal and AMS analysis

The magnetic susceptibility and the initial NRM intensity of the samples were variable, depending on the sedimentary facies. For instance, in the Boulonnais area, the samples from the Visean carbonate platform (Basse-Normandie, Moines, Leulinghen and Limont-Fontaine quarries) had negative bulk susceptibilities and very weak remanences (Fig. 5). In these cases, the calcite matrix may be the major contributor to the magnetic susceptibility, and the AMS results display a large scatter of the principal axes.

Rocks deposited in more restricted environments (black shaley limestones and siltstones) as well as the red clay levels display rather weak positive susceptibilities and relatively high NRM intensities. In these rocks, the AMS is likely to be of ferromagnetic origin. In contrast, green clay levels from the Banc Noir quarry and grey shales from the Griset quarry are characterized by relatively high positive susceptibilities and low NRM intensities (Fig. 5). This is likely to be due to a dominant paramagnetic contribution to the magnetic susceptibility from the clay minerals.

A significant mean AMS ellipsoid could only be determined for the sites with a positive magnetic susceptibility. The anisotropy parameter, P (K_{max}/K_{min}), depends on the lithology and the tectonic context but never exceeds 1.05 (maximum value in the Vireux quarry). The low values indicate that the rocks have suffered little internal deformation, and consequently, the direction of remanent magnetization is not biased. Magnetic fabrics are generally triaxial with a significant component of tectonic-induced linearity (Fig. 6a). Within localities, the anisotropy parameters and the shape of the AMS ellipsoid may vary strongly. This feature is particularly clear within the Bettrechies syncline. There, the degree of anisotropy and the planar/linear tendency of the fabric vary with the sedimentary facies (limestones exhibit dominantly prolate AMS ellipsoid, whereas shales show more planar fabrics). The general anticorrelation between the magnetic susceptibility and the anisotropy degree (Fig. 6b) may be explained by the variable amount of secondary, post-deformational, weakly anisotropic pyrite observed under the microscope.

In the majority of sites, the minimum susceptibility axes cluster close to the bedding pole, suggesting an inherited sedimentary fabrics (see Fig. 7c). In the sites from the Bettrechies syncline and from the Givet area, some minimum susceptibility axes deviate from the bedding pole and define a subtle girdle-like distribution around the maximum susceptibility axes (Fig. 7a, b, and d). The latter are generally well clustered within the bedding, defining a magnetic lineation that, in most cases, can be related to the observed structural markers.

The relations of anisotropy lineations and structures are basically of two types. The first type is characterized by lineation perpendicular to the shortening, like in the sites from Tournai, Bettrechies, Vireux and the southern part of the Griset quarry. Such a magnetic fabric, with an intersection-type lineation within the bedding, has been frequently observed in folded rock sequences

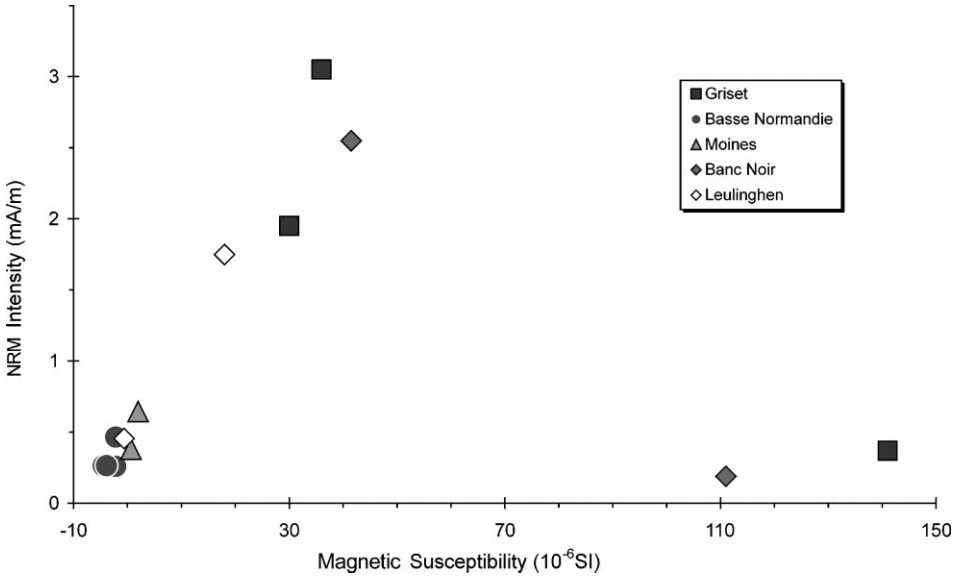


Fig. 5. Correlation diagram between the intensity of Natural Remanent Magnetization/Magnetic Susceptibility in the Boulonnais area.

(see Borradaile and Tarling, 1981; Kligfield et al., 1981; Kissel et al., 1986; Averbuch et al., 1995, for example) and can be interpreted as a composite feature due to the superposition of the tectonic fabric upon the sedimentary fabric. It is interesting to note that in Bettrechies, the magnetic lineation is normal to the late Variscan direction of shortening (NE–SW; see Fig. 7a and b) that postdates

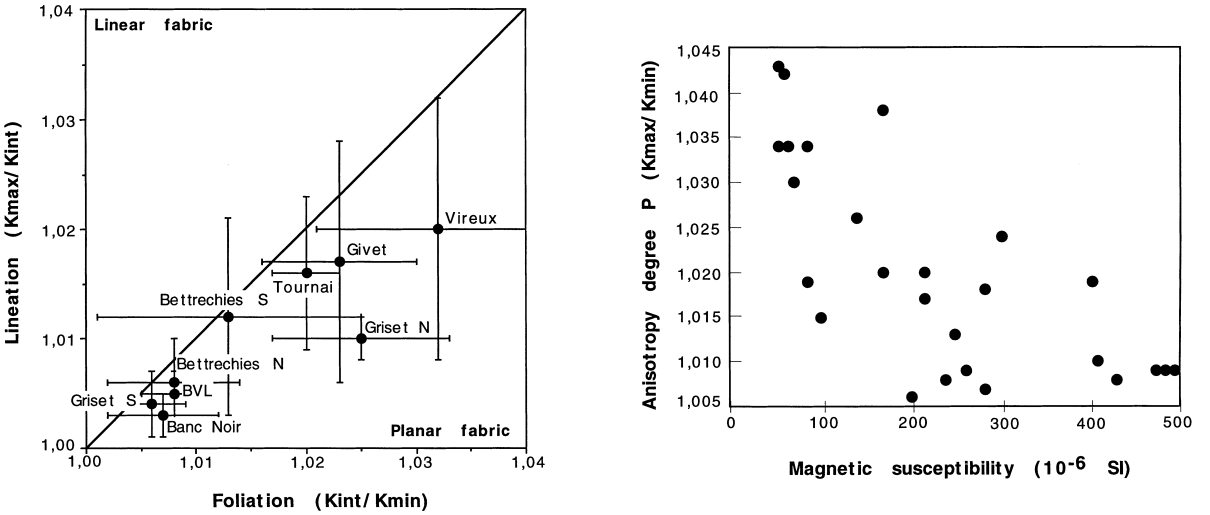


Fig. 6. Magnetic fabric parameters within the study area. (a) Diagram reporting the mean values of magnetic lineation, L (K_{max}/K_{int}) versus magnetic foliation, F (K_{int}/K_{min}). Site means are arithmetic means. Bars represent standard deviation on each parameter. The line $L=F$ separates an upper domain of dominantly linear magnetic fabrics ($L>F$) from a lower domain of dominantly planar magnetic fabrics ($F>L$). BVL: Boulonnais Visean Limestones. (b) Diagram showing the degree of anisotropy, P (K_{max}/K_{min}) versus the mean magnetic susceptibility for all samples from the Bettrechies quarry. Note the general anticorrelation of these parameters due to the influence of weakly anisotropic paramagnetic pyrite.

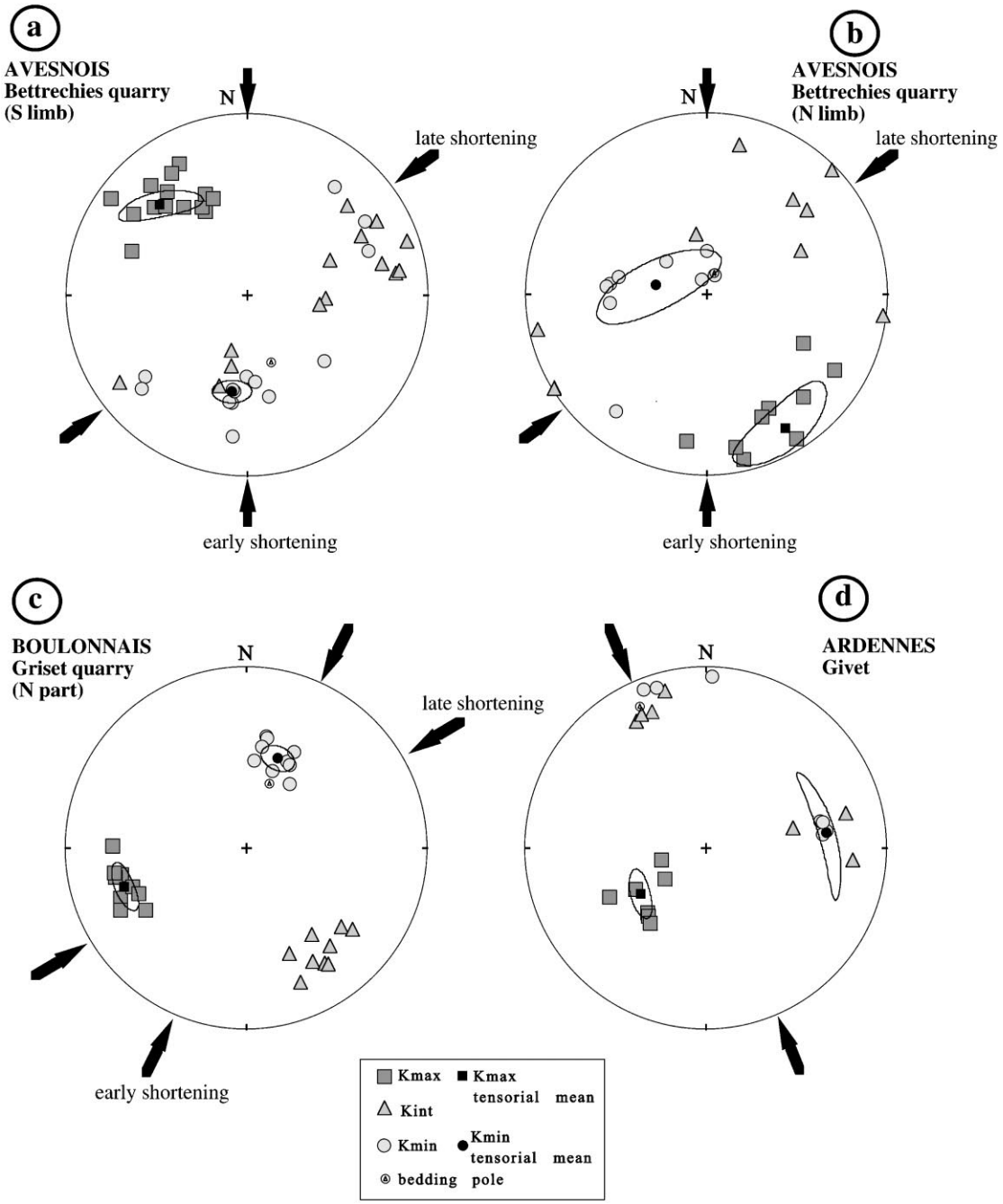


Fig. 7. Stereoplots (lower hemisphere, equal area) of representative magnetic fabrics from the area. The ellipses represent the confidence angle around the mean direction of each principal susceptibility axis.

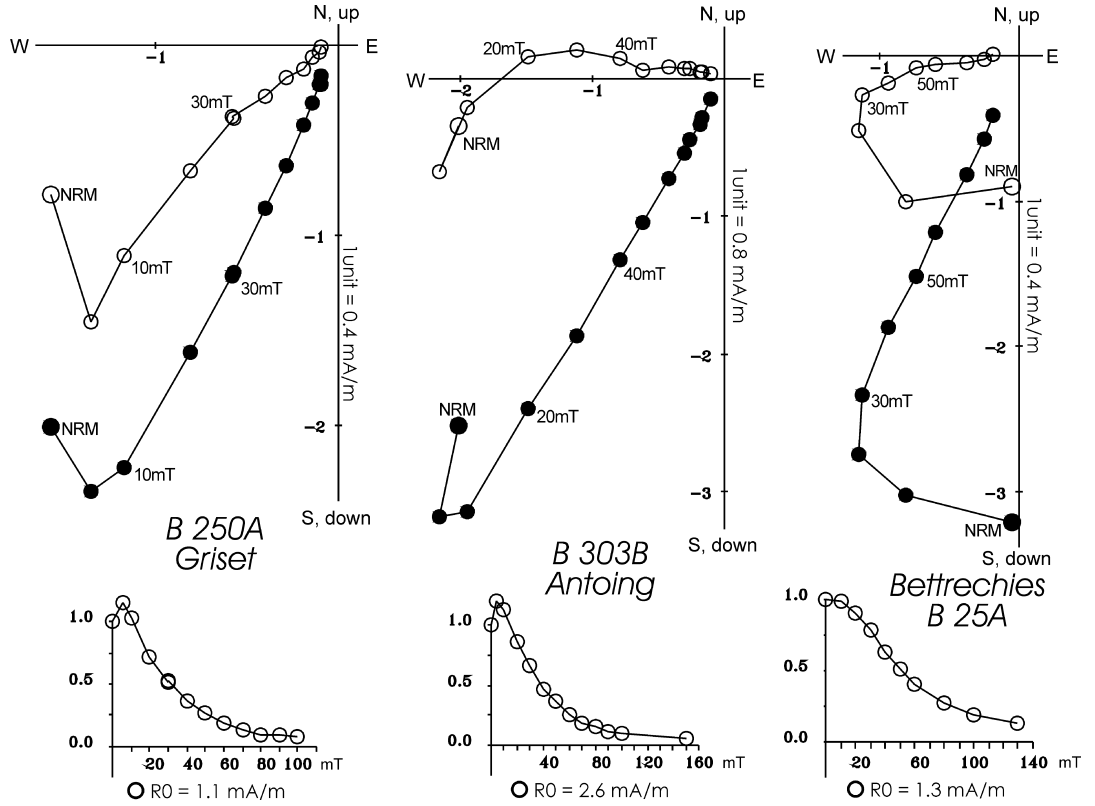


Fig. 8. Typical demagnetization curves for soft remanence. Upper row: Zijderveld diagrams. Open/solid symbols are projections on the vertical/horizontal plane. Lower row: normalized intensity versus demagnetizing field diagrams.

the major folding. A late reorganization of the fabric is also suggested by the formation of late pyrite. The second type of relationship between lineation and structures is observed at the northern part of the Griset and in the Banc Noir quarries. Here, the magnetic lineation strikes NNE–SSW to ENE–WSW (Fig. 7c), parallel to the tectonic transport directions evidenced in the Boulonnais (Averbuch and Mansy, 1998). Although an orientation of the magnetic lineation of depositional origin cannot be excluded, it seems more reasonable to interpret it as a tectonic transport-parallel lineation related to layer-parallel shear of the sedimentary sequence. The mechanisms for such a preferred orientation pattern still remain to be understood.

The site of Givet is a kind of exception, in which a well-developed lineation cannot be readily related to structural features (Fig. 7d).

4.2. Magnetic mineralogy and behaviour of NRM on demagnetization

The lithologic groups defined above were characterized by different magnetic signal intensities and also behaved differently during demagnetization.

The dark grey marls of Griset, Antoing and Bettrechies (Fig. 8) were completely demagnetized in AF fields of a maximum of 200 mT, while the other rock types needed thermal treatment for defining the components of the NRM. The positive response of the dark grey marls to AF treatment was very welcome since, during thermal treatment, above 400°C, their susceptibility rapidly increased as a result of newly formed magnetite, thus making the measurement of the NRM difficult. The softness of the magnetic material in the dark grey marls of the above mentioned three localities is corroborated

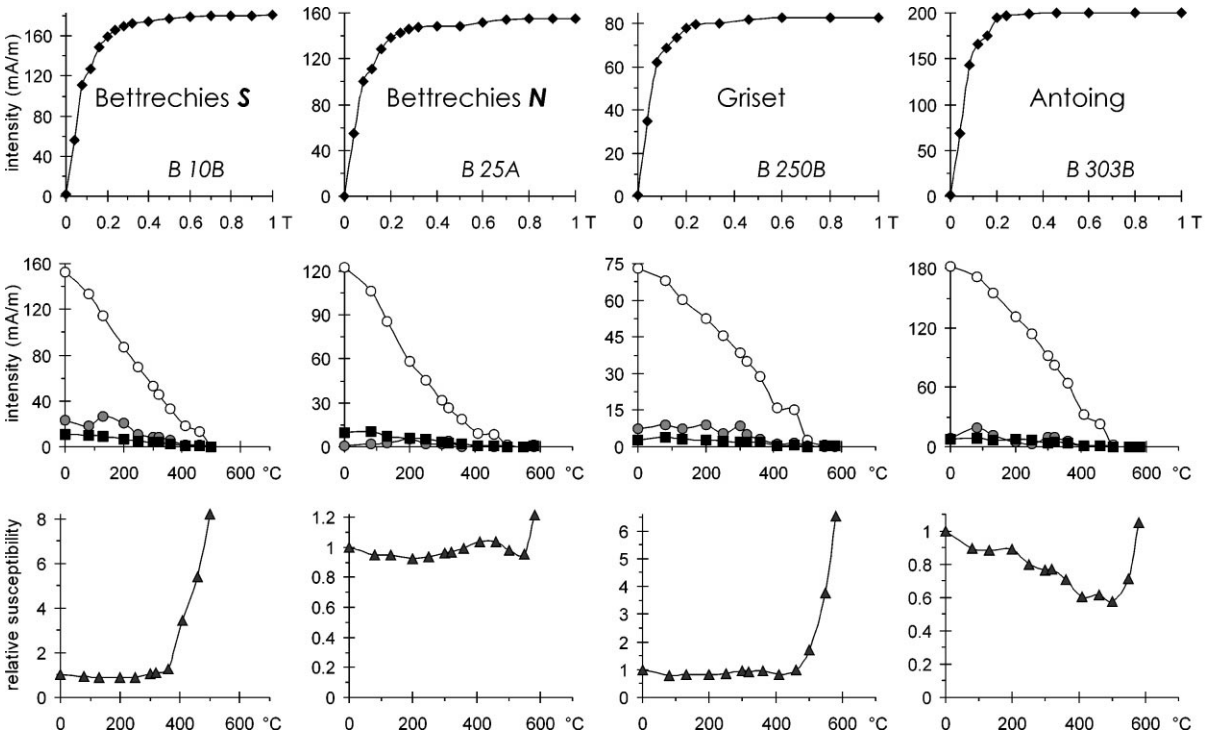


Fig. 9. Identification of magnetic minerals. First row: isothermal remanence (IRM) acquisition characteristics. Second row: thermal demagnetization of the three-component IRM; intensity versus temperature diagrams. Open symbols represent the softest component, dots the medium hard component, and squares the hardest component. Third row: behaviour of the magnetic susceptibility on heating.

by the easy saturation of the IRM acquisition curves (Fig. 9). Stepwise thermal demagnetization of specimens with dominant soft magnetic minerals eliminates the softest component (acquired in a field of 200 mT) of the three-component IRM by 500°C. The demagnetization curves also exhibit an inflexion at about 400°C, suggesting that ferrimagnetic iron sulphides and magnetite together may be present in these specimens. The presence of iron sulphide is further supported by increasing susceptibility from 400°C on. The medium hard (acquired in a field of 360 mT) and hardest (acquired in a field of 1.0 T) components in these specimens are really subordinate. They exhibit a noisy behaviour on thermal demagnetization, yet, in some cases, the trend in their decay clearly corresponds to that of the softest component (e.g., B10b, B303b, B250b in Fig. 9).

For the other rock types, typical thermal demagnetization curves are shown in Fig. 10. For them, saturation of the IRM is not achieved at 1 T, indicating the general high coercitive nature of the magnetic fraction. A subordinate soft magnetic fraction is locally associated with the dominant hard magnetic mineral (for example in Limont-Fontaine). The hard and dominant magnetic mineral is haematite, as shown by the three-axis and classical IRM thermal demagnetization curves (Fig. 11). Exceptions occur in the Moines quarry, where a drastic decrease of the IRM is observed at about 100°C, indicating the existence of goethite (B 52a in Fig. 11).

As a result of the analysis of the stepwise thermal or AF demagnetization of the NRM, the components of the NRM were defined for each specimen (Kent et al., 1983), and the site and locality palaeomagnetic mean directions were calculated (Tables 2 and 3). Most of the NRM demagnetization curves, apart from a small, recently acquired overprint component, exhibit single component NRM. In Limont-Fontaine and

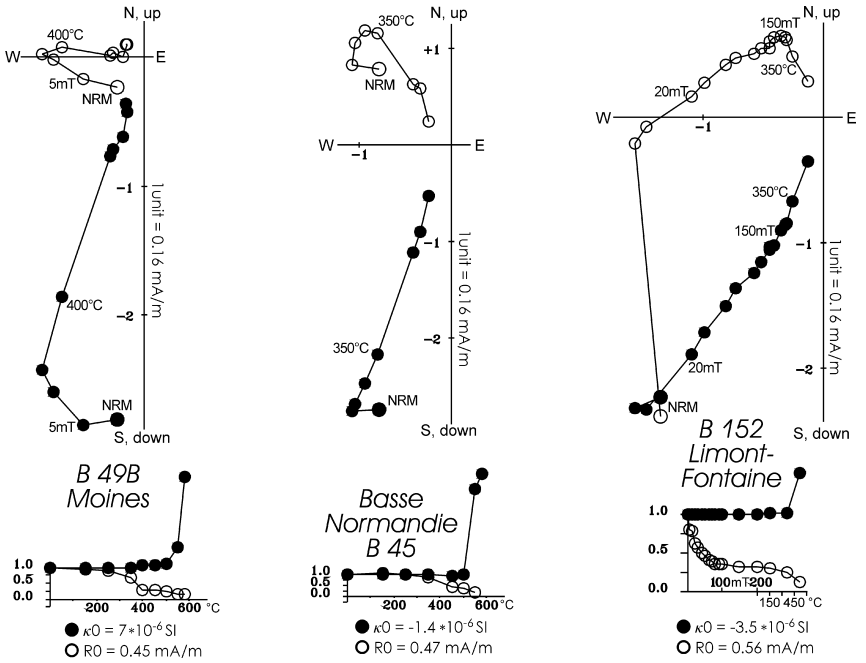


Fig. 10. Typical demagnetization curves for hard remanence. Upper row: Zijderveld diagrams. Open/solid symbols are projections on the vertical/horizontal plane. Lower row: intensity (circles) and susceptibility (dots) versus temperature curves.

Basse Normandie, both low- and a high-blocking NRM components were recognized occasionally.

Concerning the actual carriers of the remanence, there is a good agreement between the decay of the NRM and that of the three component IRM on thermal demagnetization, except in the case of the Moines quarry, where the goethite is not indicated by the decay of the NRM on heating.

5. Assessment of the palaeomagnetic directions

Several samples from each locality yielded characteristic remanent magnetization (ChRM), which is suitable for tectonic interpretation (Tables 2 and 3). The age of the acquisition of the ChRM is not, however, directly related to the stratigraphic age of the rocks.

In three cases, it was possible to sample folded structures or strata with different tilts, so that fold- or tilt-tests could be carried out on a locality level. For two such localities (Limont-Fontaine and Griset), the test is definitely negative, so the acquisition of magnetization must post-date the folding event. Therefore, these palaeomagnetic directions will be interpreted without tilt correction. For Bettrechies, partial unfolding provides the most consistent direction. This remanent magnetic signal is thus interpreted to be related to a remagnetization process acquired during a late syn-folding event.

For the other sites, which are either subhorizontal or exhibit a uniform bedding attitude within each locality, there are few arguments in favour of, or against, the pre-tectonic age of the magnetizations, except the palaeolatitude indicated by them. This aspect of the ChRM directions will be discussed later. However, it is possible to reject a pre-tectonic age for the ChRMs in the Banc Noir, Leulinghen and Moines sites in the Boulonnais due to their oversteepened negative inclinations in bedding corrected coordinates that would imply too high northern palaeolatitudes in pre-Westphalian times. As a consequence, the hard component of NRM measured within the Griset site exhibits a similar direction to that defined in the red clay samples from the Banc Noir quarry (close to the Griset quarry), is suggested to have

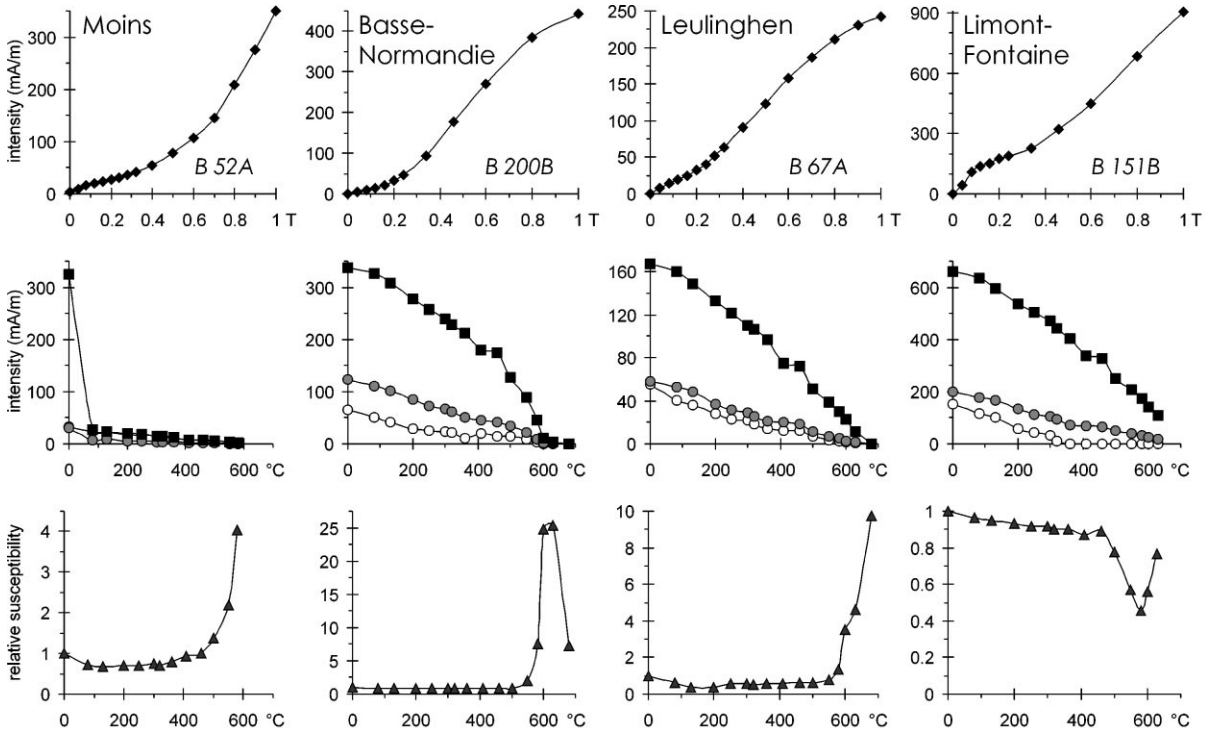


Fig. 11. IRM acquisition curves for slowly saturating samples and thermal demagnetization of the three-component IRM. Explanation as in Fig. 9.

also a post-tectonic origin. Such interpretation may be extrapolated to the extremely hard components of the NRM observed at Limont Fontaine and Basse Normandie sites.

In the Vireux and Givet sites in the Meuse valley area, where the sampled strata are tilted by the same angle within each locality, the ChRM directions are quite similar before, and widely different after, tilt correction. It is also known that these localities represent an area where metamorphic temperatures may have reached 300°C. During thermal demagnetization, the NRM signal is lost by 400°C. All these point to the post-tectonic age of the NRM in both localities.

6. Discussion of the palaeomagnetic data

6.1. Discussion of inclination data

Van der Voo (1993) compiled palaeomagnetic data for Palaeozoic time and plotted a palaeolatitude-versus-time curve for Variscan Europe. This suggests that Variscan Europe travelled from high southerly to very low southerly latitudes from Silurian through Late Devonian time. After the Late Devonian, the movement seemed to slow down, and the equator was crossed by the Armorican Massif during the Carboniferous.

Similar figures were constructed for the study area north of the Armorican Massif. As a reference framework for the new results, the expected palaeolatitudes were calculated from North European poles published by Van der Voo (1993) (his table 5.1) for several time intervals in the Palaeozoic. The general tendency shown by the black bars is that all expected palaeolatitudes are southerly until the culmination of the Variscan orogeny, followed by a sudden shift to the N during the orogeny. Afterwards, the movement to the N continued until the area probably reached 20°N by the end of the Permian (Fig. 12).

The palaeolatitudes obtained for the different components of our sites in the parautochthon

Table 2
Palaeomagnetic results from the Parautochthon[a]

Locality	n/no	D°	I°	k	$\alpha_{95}{}^\circ$	$D_C{}^\circ$	$I_C{}^\circ$	k	$\alpha_{95}{}^\circ$	Dip
Boulonnais *Ferques Unit*										
1. Griset B239–264										
B239–243 grey silty marl	5/5	206.6	+7.5	36	12.9	207.3	−23.1			200/31
B244–249 grey sandy marl	6/6	203.0	+12.1	31	12.2	203.2	−18.7			200/31
B250–253 dark grey clay	4/4	200.3	+16.8	81	10.3	200.1	−1.1			198/18
B254–257 black marl	4/4	200.7	+14.2	221	6.2	200.5	−3.7			198/18
B258–264 black shale	7/7	202.0	+17.2	1006	1.9	201.8	−3.8			201/21
Griset locality mean	26/26	202.7	+13.7	63	3.6	202.7	−10.4	38.1	4.7	
2. Banc Noir B265–275										
B271–275 purple marl component C	4/5	197.5	−22.6	62	12.0	196.0	−45.5			205/23
3. Leulinghen B59–67										
B59–61 mix of dolomite and red clay	3/3	196.9	−1.4	142	10.4	190.8	−25.1			191/24
B62–64 red clay	3/3	199.1	−10.5	104	12.1	201.2	−34.0			191/24
B65–67 dark grey dolomite	3/3	193.4	−6.3	241	8.0	194.2	−30.2			191/24
Leulinghen locality mean	9/9	196.5	−6.1	118	4.8	197.8	−29.8			
Boulonnais *Haut-Banc Unit*										
4. Basse Normandie B36–48, B199–207										
B36–40 pinkish grey limestone	5/5	181.9	+15.3	51	11	183.1	+8.5			227/9
B41–44 red limestone	3/4	169.9	+12.2	54	17	177.0	+6.1			227/9
B45–48 silty limestone	4/4	187.8	+23.8	62	12	190.5	+21.4			262/7
B199–204 grey limestone	4/6	209.5	+31.4	31	17	210.4	+23.6			226/8
B205–207 grey limestone	3/3	195.8	+24.5	26	25	297.2	+17.3			226/8
Basse Normandie locality mean	19/22	189.6	+21.8	23	7	191.3	+15.7	22	7	
Basse Normandie component C	3	202.1	−33.6	221	8	199.2	−40	113	12	
5. Moines B49–58										
B49–51, sandstone	3/3	192.6	−2.1	477	6	184.2	−40.3			220/43
B52–58 limestone	6/6	195.8	+27.3	88	6	197.8	−13.2			220/43
Tournaisis										
6. Antoing B303–314										
B303–308 black marl	6/6	210.3	−5.4	643	3	210.1	−5.6			295/2
B309–314 black marl	4/6	207.9	−3.3	208	6	207.7	−3.5			295/2
Antoing locality mean	10/12	209.3	−4.5	359	3	209.2	−4.8			

[a] n/no: number of used/collected samples D°, I°: palaeomagnetic declination and inclination before tilt correction. $D_C{}^\circ$, $I_C{}^\circ$: palaeomagnetic declination and inclination after tilt correction. k and $\alpha 95^\circ$ are Fisher's (1953) precision parameter and the radius of the 95% cone of confidence, respectively.

(Fig. 12a) and the allochthon (Fig. 12b) are plotted according to the stratigraphic age of the rock. It is clear from these diagrams that some of the components define southerly latitudes (Component A), others show latitudes close to the equator (Component B) and the rest indicate northerly latitudes (Component C). This marked difference in palaeolatitudes of the components allows an age to be assigned to the acquisition of the different magnetic components in comparison with the reference palaeolatitudes.

In two cases (1A, 4A, on Fig. 12a), the age assigned by the palaeolatitude is in accordance with the stratigraphic age, yet a negative fold test indicates that one of them (Griset) is not primary. Nevertheless, the expected palaeolatitude remains almost identical for a long period, during which remagnetization could have occurred. In the rest of the components, the acquisition clearly postdates deposition. Most of them must have been acquired during the orogeny (Components B), and the rest (Components C) well after the orogeny.

Table 3
Palaeomagnetic results from the Allochthon[a]

Locality	n/no	D°	I°	k	$\alpha_{95}{}^\circ$	$D_C{}^\circ$	$I_C{}^\circ$	k	$\alpha_{95}{}^\circ$	Dip
Avesnois										
7. Bettrechies B10–35										
B10–13 S. limb, limestone	3/3	208.2	−7.9	377	4.7	209.4	+13.5			344/31
B14–18 S. limb, shale	5/5	210.0	−7.2	1251	2.2	211.3	+15.5			344/31
B19–23 S. limb, limestone	5/5	211.6	−10.5	350	4.1	211.4	+9.9			344/31
S. limb mean	14/14	210.1	−8.6	425	1.9	211.8	+12.3			
B24–26 N. limb, limestone	3/3	204.8	+4.0	355	6.6	205.0	−11.8			197/16
B27–28 N. limb, limestone	2/2	202.9	−0.5			203.4	−16.4			197/16
B29–32 N. limb, shale	4/4	205.2	0.0	166	7.1	205.5	−15.7			197/16
B33–35 N. limb, limestone	3/3	205.4	−0.2	1235	3.5	205.9	−15.9			197/16
N. limb, mean	12/12	204.8	+0.9	334	2.4	205.2	−14.9			
Bettrechies locality mean	26/26	207.6	−4.2	140	2.4	208.2	−0.3	37	5.2	
Bettrechies[b] locality mean	7/7	206.9	−3.1	170	4.6	206.6	−3.2	590	2.6	
Avesnois										
8. Limont Fontaine B147–157										
B147–151	5/5	195.1	+1.9	39	12.4	224.0	+34.1			11/57
B152–157	5/6	213.6	+15.2	68	9.4	200.7	−58.3			170/72
Limont Fontaine locality mean	10/11	204.3	+8.7	26	9.7	195.3	+12.1	2.7	36.5	
Limont Fontaine component C	4	199.8	−33.8	113	8.7	229.0	−26.0	1.8		
Ardennes										
9. Vireux B108–116										
B108–110, 111–116	7/9	196.2	−27.4	18	14.5	190.8	+4.8	21	13.6	336/44 342/36
10. Givet B117–124										
B117–120, 121–124	5/8	193.1	−15.5	25	15.7	237.1	+51.7			155/75 overturned

[a] Variables as for Table 2.
[b] Palaeomagnetic direction at 26% unfolding.

This latter component is invariably resides in haematite.

Since there is a general tendency of a northward shift with age, we might interpret the slight differences between palaeolatitudes of components B as a slightly older magnetization in the Allochthon and a younger magnetization in the Autochthon. If the magnetization were due to tectonic events, the above sequence of remagnetization would also correspond to the known sequence of tectonic evolution (Mansy and Meilliez, 1993).

Comparing the reference palaeolatitudes (Van der Voo, 1993) and those calculated from the present study to previously published data sets from the Namur-Brabant and Ardenne areas (Molina Garza and Zijderveld 1996; shaded stripes on Fig. 12), it can be concluded that in the new data set, there seem to be magnetizations that are older, and also younger, than previously published values.

6.2. Discussion of declination data

The expected declinations in different times are plotted (Fig. 13) for the study area from data by Van der Voo (1993) together with the new declinations. The diagram suggests a broad agreement between the two data sets, since the declinations fall between 190 and 210°, but on closer examination, the declinations for group C are all far from the declinations expected from the stratigraphic age of the rocks. If these directions are moved to their estimated age of remagnetization (based on the palaeolatitudes), the declinations come quite close to those expected. The declinations for age group B display a relative scattering with sites

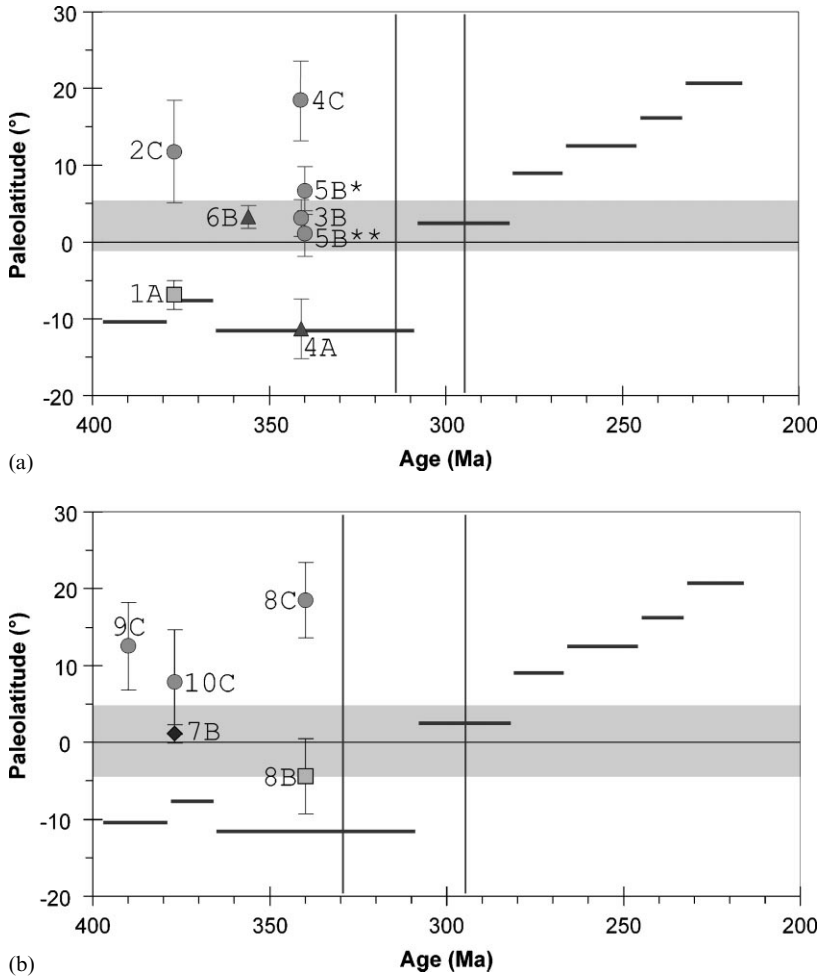


Fig. 12. Comparison of palaeolatitudes from new data with those for stable Europe. (a) Palaeolatitudes of the Parautochthon. (b) Palaeolatitudes of the Allochthon. Expected palaeolatitudes (Van der Voo, 1993) are horizontal bars; numbers refer to the localities of the present study, upper-case letters to the components of different ages. Squares are for localities with post-folding remanence, triangles for those with subhorizontal tilt, and diamonds for synfolding remanence. Dots represent localities, where, applying the tilt correction, the palaeolatitude would be too high for a Palaeozoic age of aquisition of the remanence (i.e., unrealistic). Vertical lines are the time limits for the Variscan compressive deformation in the corresponding area. The grey zone represents the interval defined by the mean palaeolatitudes of the low- and high-temperature components identified by Molina Garza and Zijderveld (1996).

showing directions close to those expected for the age of remagnetization and others closer to that expected for the stratigraphic age [for example, Antoing (6B), Bettrechies (7B) and Limont-fontaine (8B)]. Despite these small deviations from the expected declinations for the estimated age of the aquisition of the remanence, neither systematic rotation of the measured parts with respect to the reference declinations nor any difference along the studied belt can be found in these data. It is therefore concluded that the differences in declination are best explained by a moderate scatter around the mean value of 200° both in the Parautochthon and in the Allochthon, partly due to the different ages of remagnetizations.

The earliest remagnetizations (age group A) show interesting results: the Griset soft component (1A in Fig. 13a) has declinations that fall near to

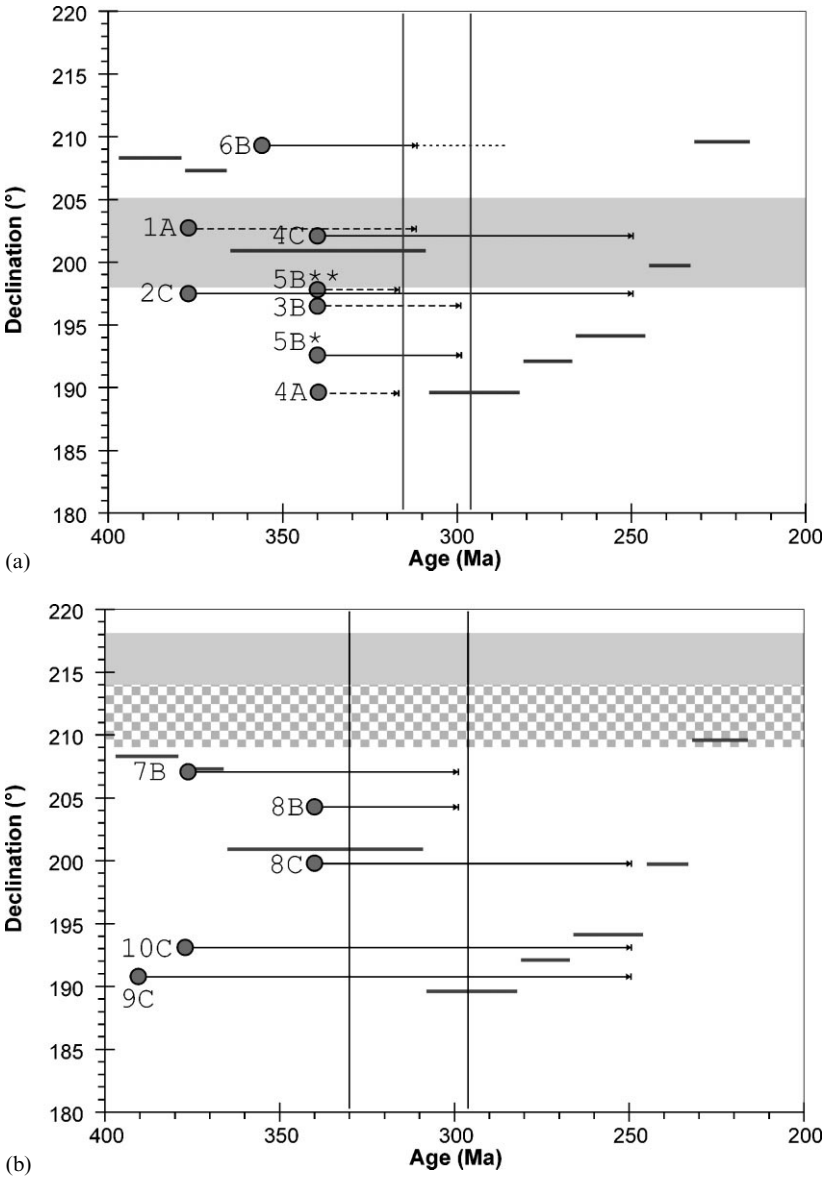


Fig. 13. Comparison of palaeodeclinations from new data with those for stable Europe. (a) Palaeodeclinations of the Paraautochthon. (b) Palaeodeclinations of the Allochthon. Expected palaeodeclinations (Van der Voo, 1993) are horizontal bars; numbers refer to the localities of the present study, upper-case letters to the component of different ages. Light and broken horizontal lines extend from the stratigraphic ages to the estimated ages of the acquisition of the remanence. Vertical lines are the time limits for the Variscan compressive deformation in the corresponding area. The grey zone represents the interval defined by the mean palaeodeclinations of the low- and high-temperature components identified by Molina Garza and Zijderveld (1996). In the Parautochthon, the declinations of the low-temperature components fall between 181 and 208°, and that of the high-temperature component between 179 and 216°. The same values for the allochthon are 211–230 and 210–234°, respectively. From the data set by Thominski et al. (1993), the mean declinations for the lower Devonian through lower Carboniferous age groups define the checked interval.

the expected declinations, suggesting litte rotation. The soft component of Basse Normandie (4A in Fig. 13a), however, shows a difference of 10–15° from the expected declination. This locality is in the same area as Griset, but in an upper tectonic slice, at the footwall of the Midi Fault. The confidence circles of the two palaeomagnetic directions overlap, i.e., the two declinations do not differ in a significant way. Nevertheless, they might be regarded as indicating a 10–15° counterclockwise rotation of Basse Normandie with respect to the non-rotating Griset and reference frame.

Assessment of previously published declination data is somewhat difficult because different authors suggested different remagnetization ages and published different declination values for the same Ardenne area. Molina Garza and Zijderveld (1996) suggested a 15° clockwise rotation of the Ardenne Allochthon with respect to the Brabant Parautochthon, which was recorded in both their high- and low-temperature components. This rotation is thus younger than 300 Ma. All the other authors interpreted slightly larger (20–30°) clockwise rotations in both their earlier and later remagnetizations.

In contrast, Givet and Vireux localities (component C) of the present study, in the same Ardenne region, show no significant rotation. This contradiction is even more interesting if we note that in the Parautochthon, the new declination data scatter around those of the mean declinations of the high- and low-temperature components of Molina Garza and Zijderveld (1996), i.e the older and new data sets are not different (Fig. 13).

In conclusion, the declination data obtained here do not indicate any significant rotation along the central and western portions of the Variscan chain of Southern Belgium–Northern France with respect to its foreland. Therefore, it is reasonable to compute palaeomagnetic pole positions from the overall mean directions of component A, B, and C, respectively (Table 4). There is perhaps one exception. The small (10–15°) counterclockwise deviation of the declination of the upper tectonic slices of the Boulonnais (Brabant Parautochthon) with respect to the foreland may have some tectonic significance.

7. Discussion on geodynamic models

The Variscan front of NW Europe exhibits a progression of deformation events. First, the Southern part of the Allochthon was deformed in Late Visean time with folding, cleavage formation and thrust faulting progressing northward during Late Carboniferous time. Fold axes are generally oriented NE–SW in the eastern Ardenne, and E–W in the central Avesnois. This deformation is accompanied in the Parautochthon by the formation of a foreland basin where thick molasse deposits (coal measures) accummulated. Folding in the middle part of the Parautochthon appears slightly later than Westphalian D. Fold axes in the Parautochthon generally follow the strike of the deformation front and fold axes in the Allochthon. In the Boulonnais (part of the Parautochthon), fold axes strike NW–SE. The structural directions of the Variscan front thus define an arc concave towards the north.

Table 4
Palaeomagnetic overall-mean directions and pole positions for components of early Variscan (A), main Variscan (B) and post-Variscan, probably Permian (C) age[a]

	N	D°	I°	k	$\alpha_{95}{}^\circ$	Pole		δp	δm
						Longitude	Latitude		
Component A	10	195.5	+17.8	35	8.2	166	29	4.4	8.5
Component B	16	203.4	−2.9	72	4.4	154	37	2.2	4.4
Component C	5	197.6	−26.6	94	7.9	154	51	4.6	8.6

[a] N: number of sites. D°, I°: palaeomagnetic declination and inclination in the geographic system. k, $\alpha_{95}{}^\circ$ as in Table 2. δp and δm are the semi-axes of the 95% confidence ellipse associated with the mean poles (Fisher, 1953).

The main Midi Fault is an out-of-sequence fault that cuts folds, implying Westphalian C, as well as cutting faults of the Parautochthon, and has a variable offset along its strike. Near Aachen, the displacement does not exceed a few kilometres, in the central part of the arc (near Bettrechies) it is at least 80 km (Mansy et al., 1997), and in the western segment of the arc (Boulonnais-Artois), the displacement is at least 15–20 km, while in southern England it is not known.

Several models possibly explain the present arcuate shape of the Variscan front of Southern Belgium–Northern France. The first (oroclinal bending; Fig. 14a) implies a wedge-shaped obstacle or promontory at the Brabant margin, on which a linear orogen is moulded. In this model the originally parallel, linear structures are rotated as the allochthon moves northward until they become (grosso modo) parallel with the margins of the obstacle. This model would imply clockwise and counterclockwise rotation of declination directions on the western and eastern branches of the chain, respectively. The amount of rotation would depend on the angular difference between the linear chain and the margins of the obstacle.

The second model (inherited shapes, Fig. 14b)

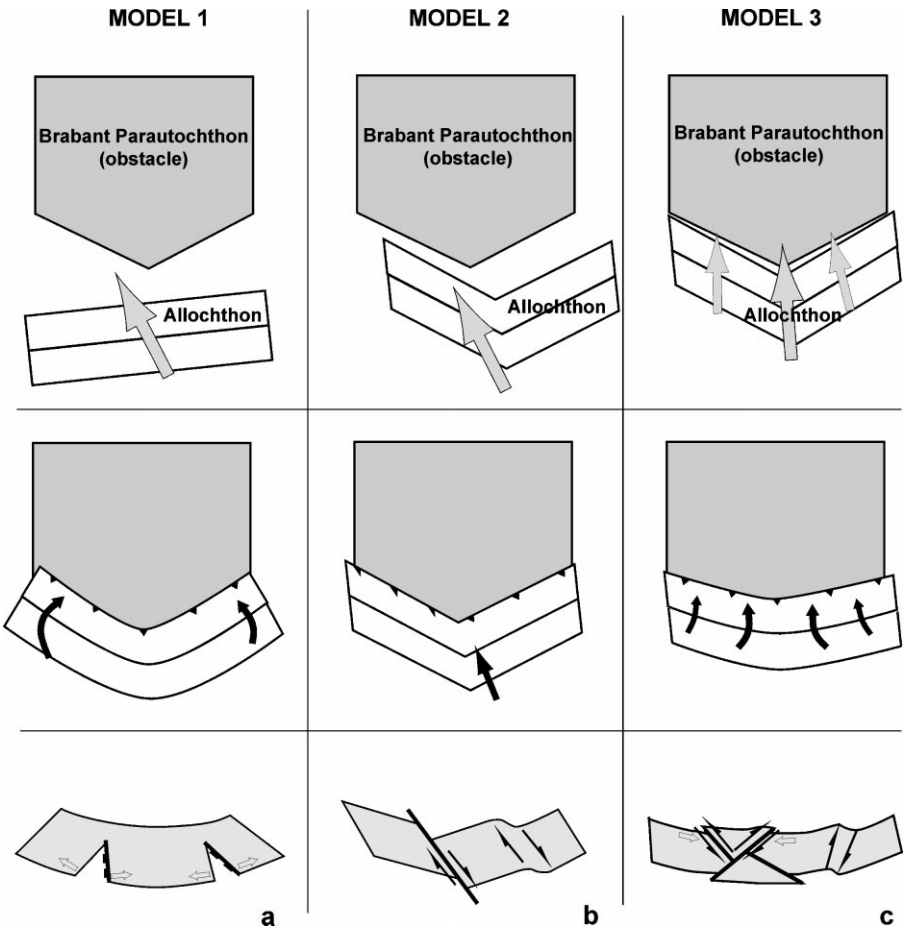


Fig. 14. Possible geodynamic models explaining the curvature of the Northern France–Southern Belgium Variscan thrust front. (a) Moulding of an originally linear orogen. (b) Collision of an originally arched orogen. (c) Unfolding of an originally arched orogen with the help of a variable offset thrust fault. Upper row: before orogeny. Arrows indicate future movements. Middle row: after orogeny. Black arrows indicate induced rotations. Lower row: sketch of internal deformations induced by the models.

presumes an originally arcuate shape of the Allochthon, which has the shape of the obstacle. When deformed, these would produce structures parallel to the margin of the Brabant obstacle, regardless of the tectonic transport directions. In this case, the structures were originally arranged in an arcuate pattern with no significant rotation in either branch of the chain. During progressive deformation, the shape of the original arc may be accommodated by strike-slip fault zones according to local space problems. A limited amount of rotation is possible along these zones.

In the third model (Fig. 14c), an original arcuate shape of the arc is again postulated, similar to the second model, but with the curvature diminishing with time. The decrease in curvature is accommodated by thrust-faulting, the offset of which varies from nil at both ends to maximal in the central part. The thrust-faulting would imply opposed, clockwise and counterclockwise rotations in the eastern and western branches, respectively. In models 1 and 3, the induced rotations imply orogen-parallel elongation or shortening. These deformations may be accommodated by structures at a high angle to the orogenic front.

The presently available palaeomagnetic data do not support the first model. The data set of the present study shows no significant rotation either in the time interval when the components of different ages (age groups A, B, C) were acquired, or along the chain, though age group B data correspond in age to those of Thominski et al. (1993) and Molina Garza and Zijderveld (1996). Arc forming would be best seen with the help of pre-tectonic declination directions. In the eastern Ardenne sector, there are no such data so far, as the only available declinations are syn- to post-tectonic. In the western sector, however, early tectonic declinations (Griset, Basse Normandie) are present and do not show any rotation.

The second model explains rather well the lack of significant palaeomagnetic rotations, and it needs only secondary deformation zones to explain the measured small rotations. The Meuse valley segment, measured by Molina Garza and Zijderveld (1996), is interpreted as a NW–SE-oriented dextral shear zone, creating the observed 15° clockwise rotation (Fig. 1). Here, near Givet, axial planes of folds are also rotated, cleavage is tilted and folds are refolded (Lacquement et al., 1997). Other smaller rotations would be also explained by local, late, accommodating strike-slip zones.

Model 3 could explain the clockwise rotation of the eastern Ardenne segment, together with the increased offsets observed along the Midi Fault from the east towards the central part. A symmetrical model would imply a small counterclockwise rotation in the western, Boulonnais segment and a decrease in the offsets of the Midi Fault from the center towards the west as well. The more daring interpretation of the Boulonnais palaeomagnetic data could support some limited counterclockwise rotation, but there is no information on the amount of offset along the western, covered segments of the Midi Fault. In this model, the orogen-parallel shortening could be accommodated by conjugate sets of strike-slip faults, along which material would be extruded to the north or south.

Acknowledgements

This work was partly sponsored by a Hungarian–French intergovernmental project (Balaton F-49/95) and by the Hungarian Science Research Found (OTKA, No. T029805). We are grateful to Prof. D.H. Tarling and an anonymous reviewer for useful comments. F. Bergerat from Université Paris VI. (France) and S. Vandycke from Mons University (Belgium) are thanked for field assistance and organization. The quarry owners and staff in the Boulonnais, Tournai area and Avesnois are thanked for their kind cooperation and local guidance.

References

Ahrendt, H., Clauer, N., Hunziker, J.C., Weber, K., 1983. Migration of folding and metamorphism in the Rheinische Schiefergebirge deduced from K–Ar and Rb–Sr age determinations. In: Martin, H., Eder, F.W. (Eds.), Intracontinental Fold Belts. Springer, Berlin, pp. 323–338.

Averbuch, O., Mattei, M., Kissel, C., Frizon de Lamotte, D., Speranza, F., 1995. Cinématique des déformations au sein

d'un système chevauchant aveugle: l'exemple de la Montagna dei Fiori (front des Apennins centraux, Italie). Bull. Soc. géol. Fr. 166, 451–461.

Averbuch, O., Mansy, J.-L., 1998. The 'Basse-Normandie' duplex (Boulonnais, N France): evidence for an out-of-sequence thrusting overprint. J. Struct. Geol. 20, 33–42.

Bless, M.J.M., Bouckaert, J., Paproth, E., 1980. Environmental aspects of some Pre-Permian deposits in NW Europe. Meded. Rijks Geol. Dienst 32 (1), 3–13.

Bois, Ch., Cazes, M., Choukroune, P., Gariel, O., Hirn, A., Le Gall, B., Lefort, J.P., Matte, P., Pinet, B., 1994. Seismic reflexion images of the Pre-mesozoic crust in France and adjacent areas. In: Pre Mesozoic in France and Related Areas. Springer, Berlin, pp. 3–48.

Borradaile, G.J., Tarling, D., 1981. The influence of deformation mechanisms on magnetic fabrics in weakly deformed rocks. Tectonophysics 77, 151–168.

Bouroz, A., Chalard, J., Dalinval, A., Stievenard, M., 1962. La structure du bassin houiller du Nord de la région de Douai à la frontière belge. Ann. Soc. géol. Nord LXXXI, 173–220.

Chalard, J., 1945. La faille de Vicoigne du terrain houiller du Nord de la France. Ann. Soc. géol. Nord LXV, 177–196.

Chamley, H., Proust, J.N., Mansy, J.L., Boulvain, F., 1997. Diagenetic and paleogeographic significance of clay and other sedimentary components in the middle Devonian limestones of Western Ardenne, France. Palaeogeogr., Palaeoclimatol., Palaeoecol. 129, 369–385.

Cooper, M.A., Garton, M.R., Hossack, J.R., 1983. The origin of the Basse-Normandie duplex, Boulonnais, France. J. Struct. Geol. 5, 139–152.

Edel, J.B., Coulon, M., 1986. Mise en évidence de rotations tardihercyniennes à partir d'un profil paléomagnétique à travers l'Ardenne et le Brabant. Ann. Soc. géol. Nord 105, 139–144.

Fielitz, W., Mansy, J.-L., 1999. Pre- and synorogenic burial metamorphism in the Ardenne and neighbouring areas (Rhenohercynian zone, Central European Variscides). Tectonophysics 309, 227–256.

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. Lond., Ser. A 217, 295–305.

Fourmarier, P., 1922. Tectonique générale des terrains paléozoiques de la Belgique. C.G.I., Livret-guide pour la XIII° session, Belgique. Exc. C2.

Franke, W., Engel, W., 1986. Synorogenetic sedimentation in the Variscan belt of Europe. Bull. Soc. Géol. Fr. 1 (8), 25–33.

Gosselet, J., 1888. L'Ardenne. Mém Carte Géol. Fr. Baudry, Paris. 881 pp

Hance, L., Ghysel, P., Laloux, M., DeJonghe, L., Mansy, J.L., 1999. Influence of heterogeneous layering on orogenic deformation in the Variscan front zone (eastern Belgium). Tectonophysics 309, 161–177.

Helsen, S., 1992. Conodont colour alteration maps for palaeozoic strata in Belgium, northern France and westernmost Germany. Preliminary results. Ann. Soc. géol. Belg. 115 (1), 135–143.

Helsen, S., 1995. Burial history of Palaeozoic strata in Belgium as revealed by conodont colour alteration data and thickness distributions. Geol. Rundschau 84, 738–747.

Hennebert, M., 1998. L'anticlinal faillé du Mélantois-Tournaisis fait partie d'une structure en fleur positive tardi-varisque. Ann. Soc. géol. Nord 6, 65–78.

Humblet, E., 1941. Le Bassin houiller de Liege. Revue Universelle des Mines 17 (12), 357–377.

Kenis, I., Muchez, P., Sintubin, M., Mansy, J.-L., Lacquement, F., 2000. Thermal constraints on the progressive deformation in the frontal area of the variscan Ardenne Allochthon (Bettrechies, France). J. Struct. Geol. 22, 589–602.

Kent, J.T., Briden, J.C., Mardia, K.V., 1983. Linear and planar structure in order multivariate data as applied to progressive demagnetization of paleomagnetic remanence. Geophys. J. R. Astr. Soc. 75, 593–621.

Khatir, A., 1990. Structuration et déformation progressive au front de l'allochtone ardennais (Nord de la France). Publ. Soc. Géol. Nord 18, 293 pp.

Kissel, C., Barrier, E., Laj, C., Lee, T.Q., 1986. Magnetic fabric in undeformed marine clays from compressional zones. Tectonics 5, 769–781.

Kligfield, R., Owens, W.H., Lowrie, W., 1981. Magnetic susceptibility anisotropy, strain and progressive deformation in Permian sediments from the Maritime Alps (France). Earth Planet. Sci. Lett. 55, 181–189.

Kossmat, F., 1927. Gliederung des varistischen Gebirgsbaues. Abh. Sachs. geol. Landesamt heft 1, 1–39.

Lacquement, F., Mansy, J.-L., Mercier, E., Averbuch, O., Meilliez, F., 1997. Mécanismes de plissement disharmonique dans le Paléozoique ardennais. Aardkundige Mededelingen 8, 119–122.

Lacquement, F., Mansy, J.-L., Hanot, F., Meilliez, F., 1999. Retraitement et interprétation d'un profil sismique pétrolier méridien au travers du Massif Paléozoique ardennais (Nord de la France). C. R. Acad. Sci., Paris 329, 471–477.

Laloux, M., DeJonghe, L., Ghysel, P., Hance, L., Mansy, J.L., 1997. Nouveaux concepts tectoniques à partir de la révision des cartes Fléron-Verviers et Limbourg-Eupen (Est de la Belgique). Ann. Soc. géol. Nord 5, 23–46.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.

Mansy, J.L., Meilliez, F., 1993. Eléments d'analyse structurale à partir d'exemples pris en Ardenne–Avesnois. Ann. Soc. géol. Nord 2, 45–60.

Mansy, J.-L., Meilliez, F., Mercier, E., Khatir, A., Boulvain, F., 1995. Le rôle du plissement disharmonique dans la tectogenese de l'allochtone ardennais. Bull. Soc. géol. Fr., Paris 166 (83), 295–302.

Mansy, J.L., Lacquement, F., Meilliez, F., Hanot, F., Everaerts, M., 1997. Interprétation d'un profil sismique pétrolier, sur le méridien de Valenciennes (Nord de la France). Aardk. Mededel. 8, 127–129.

Mansy, J.L., Amedro, F., Auffret, J.P., Guennoc, P., Lamarche, J., Lefevre, D., Robaszinski, F., Somme, J., Vidier, J.P., press. Carte géologique de Marquise (50000eme). second ed., BRGM, Orléans. in press.

Matte, P., 1986. Tectonics and plate tectonic model for the Variscan belt of Europe. Tectonophysics 126, 329–374.

Matthews, S.C., 1984a. Tectonic antecedence of Variscan geology in Belgium. Bull. Soc. belge géol., Bruxelles 93, 33–44.

Matthews, S.C., 1984b. Northern margins of the variscides in the North Atlantic region: comments on the tectonic context of the problem. In: Hutton, D., Sanderson, D. (Eds.), Variscan Tectonics of the Atlantic Region, Geol. Soc. Spec. Publ., pp. 71–85.

Meilliez, F., Mansy, J.L., 1990. Déformation pelliculaire différenciée dans une série lithologique hétérogene: Le Dévono-Carbonifère de l'Ardenne. Bull. Soc. géol. Fr., VI 1, 177–188.

Meilliez, F., Andre, L., Blieck, A., Fielitz, W., Goffette, O., Hance, L., Khatir, A., Mansy, J.L., Overlau, P., Verniers, J., 1991. Ardenne-Brabant. Sci. Géol. Bull. 44, 3–29.

Meilliez, F., Fielitz, W., Laduron, D., Mansy, J.L., Manby, J.L., 1994. The deep structure of the Ardennes Variscan thrust belt from structural and ECORS seismic data: Discussion. J. Struct. Geol. 16 (3), 431–432.

Molina Garza, R.S., Zijderveld, J.D.A., 1996. Paleomagnetism of Palaeozoic strata, Brabant and Ardennes Massifs, Belgium: Implications of prefolding and postfolding Late Carboniferous secondary magnetizations for European apparent polar wander. J. Geophys. Res. 101, 15799–15818.

Muchez, P., 1989. Comparison of the Visean strata north and south of the London–Brabant Massif. Bull. Soc. Belg. Géol. 98, 155–162.

Perroud, H., 1986. Paleomagnetic evidence for tectonic rotations in the Variscan mountain belt. Tectonics 5, 205–214.

Piqué, A., Huon, S., Clauer, N., 1984. La schistosité hercynienne et le métamorphisme associé dans la vallée de la Meuse, entre Charleville-Mèziéres et Namur (Ardennes franco-belges). Bull. Soc. Belge Géol. 93, 55–70.

Potdevin, J.L., Goffette, O., 1991. Les assemblages métamorphiques du filon de diabase de la Grande Commune (Massif Rocroi), des témoins d'une évolution rétrograde varisque en Ardenne. C. R. Acad. Sci. Paris 312 (2), 1545–1550.

Raoult, J.-F., Meilliez, F., 1987. The Variscan front and the Midi fault between the Channel and the Meuse River. J. Struct. Geol. 9, 473–479.

Thominski, H.P., Wohlenberg, J., Bleil, U., 1993. The remagnetization of Devono-Carboniferous sediments from the Ardenno-Rhenish Massif. Tectonophysics 225, 411–431.

Van der Voo, R., 1993. Paleomagnetism of the Atlantic Tethys and Iapetus Oceans. Cambridge University Press, Cambridge. 411 pp

Winterfeld, C.V., Walter, R., 1993. Die varisziche Deformations front des nordwestlichen Rheinischen Schiefergebirges—Ein bilanziertes geologisches Tiefenprofil über die Nordeifel. N. Jb. Geol. Paläont. Mh. 1993 (5), 305–320.